

ALEXANDRA WEBER

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Ottawa and Mississauga, Ontario, Canada

EDUCATION

B.Sc. Honours Biology

Sept 2020 – Present

Carleton University

Ottawa, ON

- CGPA: 11.36/12 (A)
- Major CGPA: 11.85/12 (A)
- Graduation date: April 2025

Minor Earth Science: Earth Resources and Processes

Sept 2021 – Present

Carleton University

Ottawa, ON

- Minor GPA: 12/12 (A+)

ACADEMIC AWARDS

Academic Scholarships

Carol Shields Scholarship

Sept 2024 - May 2025

- Awarded by Carleton University to a student who has completed credits beyond entrance requirements and has demonstrated a high potential for university studies.

E.W.R. Steacie Scholarship

Sept 2023 - May 2024

- Awarded by Carleton University for academic excellence in 2022/2023 school year.

Lester B. Pearson Award

Sept 2022 – May 2023

- Awarded by Carleton University for academic excellence in 2021/2022 school year.

Awards

Life Sciences Day 6.0

May 25th 2023

1st Place Undergraduate Poster Award

Academic Achievements

Dean's Honour List

21/22, 22/23, 23/24 Academic Years

- Distinction is given to students with a CGPA above 10.0 (A).

FUNDING RECEIVED

Undergraduate Research Grant

September 2021

I-CUREUS

- Granting agency: Carleton University and Dr. Tetsuto Miyashita (Canadian Museum of Nature)
- Funding received to complete a report on the morphological differences between wildtype and nkx3.2 mutant zebrafish.

CONFERENCE PRESENTATIONS

Canadian Society for Vertebrate Paleontology

May 2023

Poster presentation - Primary Presenter

- Poster Title: "A description of the osteological skull morphology of *Danio rerio* with the intent of using it for the comparative analysis of nkx3.2-/- mutants."

Life Sciences Day 6.0

May 2023

Poster presentation - Primary Presenter

- Poster Title: "A description of the osteological skull morphology of *Danio rerio* with the intent of using it for the comparative analysis of *nkx3.2* -/- mutants."
- Won first place for best undergraduate poster

Ottawa-Carleton Institute of Biology

April 2023

Poster presentation - Primary Presenter

- Poster Title: "A description of the osteological skull morphology of *Danio rerio* with the intent of using it for the comparative analysis of *nkx3.2* -/- mutants."

Canadian Society of Zoologists

May 2022

Poster presentation - Primary Presenter

Poster Title: "Survey of Coiling Directions in Snake Embryos"

WORK/RESEARCH EXPERIENCE

Science Lead

Sept 2023- Present

Carleton Planetary Robotics Team

Carleton University

- CPRT is an engineering design team that builds and develops a Mars rover with the intent of competing in international competitions.
- In charge of the overall design of the life detection and environmental analysis capabilities onboard our rover.

Teaching Assistant

Sept 2024- Dec 2024

BIOL1103 Foundations in Biology

Carleton University

- Helped conduct in class worksheets to help student's understanding of the material
- Marked unit tests and Final Exam

Teaching Assistant

Sept 2023- Dec 2023

BIOL3305 Human and Comparative Physiology

Carleton University

- Ran review sessions for course content before examinations
- Conducted in class quizzes to gauge students understanding of the material
- Proctored and Marked Midterm and Final Exam

Research Assistant

May - Aug 2023

Bruin Laboratory

Carleton University

Project Title: Impact of cisplatin on beta cell survival and function

- Looked at pancreas morphology using Histology and Immunostaining techniques.
- Maintained an INS1 832/3 cell line for experimental uses.

Research Affiliate

May - Aug 2022

Agriculture and Agri-Food Canada

London, ON

- Conducted laboratory and field work for 2 biological control projects, including dissections, PCR, field specimen collection, and data analysis.
- Maintained insect colonies using a strict and organized schedule.

I-CUREUS Internship

Nov 2021 - Mar 2022

Canadian Museum of Nature

Ottawa, ON

Granting agency: Carleton University and Dr. Tetsuto Miyashita (Canadian Museum of Nature)

- Segmentation of Zebrafish skeleton for both wild type specimen and mutant jawless specimen using 3D slicer software.
- Compared the morphological differences between the wild type and mutant.

- Final product was an interactive 3D anatomical atlas for both specimens.

Administrative Assistant (Seasonal)

May-Sept 2020/21

UTEX Scientific Instruments

Mississauga, ON

- Managed internal affairs, bookkeeping, purchasing, and customer care, with organization and precision. Leading to increased efficiency in day-day operations and customer satisfaction.

APPLIED PROJECTS

Research Project

Jun 2021 – Ongoing

Survey of Snake Embryo Coiling Direction

- Collecting and analyzing snake embryo specimens from all over the world. Looking to determine if an asymmetrical bias occurs between sinistral or dextral coiling of snakes during development.
- Presented results and conclusions via a poster at the Canadian Society of Zoologists annual conference.

Independent Research Project

Jan – Apr 2024

Upper Body Determinants of the Velocity and Speed of a Fencing Lunge

- Completed for Biomechanics (BIOL3501) at Carleton University
- Collected and analysed kinematic and kinetic Biomechanical data
- Presented research and findings in a poster and paper format.

Independent Research Project

Jan – Apr 2024

BPA-Induced Lipid Accumulation in HepG2 Cells and the Therapeutic Capabilities of Trigonelline.

- Completed as part of Advanced Cell Culture and Tissue Engineering (BIOL4201) at Carleton University
- Maintained HepG2 cells in culture with 4 Tx groups.
- Performed an oil red assay and lipid extraction semi-quantification.
- Presented research and findings in a 15-minute presentation with mock-thesis defense format and question period

Research Project

Aug 2024

Abiotic influences of spatial variations in phytoplankton and heterotrophic bacteria abundances across Barkley Sound.

- Addressed whether the new Bamfield Marine Science Centre (BMSC) sewage treatment plant resulted in a reduction of chlorophyll-a levels in Bamfield and Grappler Inlets and what additional abiotic factors affect the spatial distribution of heterotrophic bacteria and phytoplankton in Barkley Sound

Methods Development

May - Aug 2022

Survey of Spotted Wing Drosophila Populations in Southern Ontario

- Helped develop the methods used to survey the population distribution of Spotted Wing Drosophila in Southwestern Ontario.
- Created the system used for specimen organization and data collection.
- Contributions acknowledged in;
Garipey TD, et al. (2024) Widespread establishment of adventive populations of Leptopilina japonica (Hymenoptera, Figitidae) in North America and development of a

multiplex PCR assay to identify key parasitoids of *Drosophila suzukii*
(Diptera, Drosophilidae). NeoBiota 93: 63-90. <https://doi.org/10.3897/neobiota.93.121219>

PRACTICAL QUALIFICATIONS

Training

Community Fishers Program - CTD Operation

Ocean Networks Canada

August 2023

Bamfield, BC

Standard First Aid and CPR level "C" and AED

July 2023

Ottawa Paramedic Service Public Education

- Valid until: July 2026

PalEO Resource Workshops

Sept - Dec 2021

Paleontology and Evolution Ottawa

Ottawa, ON

- Participated in workshops on specimen imaging, photogrammetry, statistical programs, histology, cladistics, sequence analysis, and effective scientific communication.

Open Water Diver Certification

Mar 2014

Professional Association of Diving Instructors (PADI)

Skills

3D Reconstruction (3D Slicer)

Independent Research Project

Nov 2022-Apr 2025

PCR and Gel Electrophoresis

Agriculture and Agri-Food Canada

May 2022

Bruin Lab – Carleton University

May-Aug 2023

Introductory Genetics (BIOL2104)

Jan-Apr 2022

Cell Culture

Advanced Cell Culture and Tissue Engineering (BIOL4201)

Jan-Apr 2024

Bruin Lab – Carleton University

May-Aug 2023

Cell Biology (BIOL3201)

Sept-Dec 2022

Statistical Analysis

Ecology (BIOL2600)

Sept-Dec 2021

Introduction to Statistical Modeling (STAT2507)

Jan-Apr 2021

Primer Design and Validation

Molecular Genetics (BIOL3104)

Sept-Dec 2023

Bruin Lab – Carleton University

May-Aug 2023

Histology and Immunostaining

Advanced Cell Culture and Tissue Engineering (BIOL4201)

Jan-Apr 2024

Bruin Lab – Carleton University

May-Aug 2023

Cell Biology (BIOL3201)

Sept-Dec 2022

Basic Local Alignment Search Tool (BLAST)

Molecular Genetics (BIOL3104)

Sept-Dec 2023

Bruin Lab – Carleton University

May-Aug 2023

Introductory Genetics (BIOL2104)

Jan-Apr 2022

Languages

Russian

First year Russian Class (RUSS2010) - Carleton University

Sep – Dec 2023

Spanish

Second-year Spanish Class (SPAN2010) - Carleton University
High School Spanish - Oakville Trafalgar High School
German
German School - Freidrich Schiller Schule

Sept - Dec 2020
Sept 2016 - Jun 2018
Sept 2009 - Jun 2014

REFERENCES

Dr. Tetsuto Miyashita
Canadian Museum of Nature
Email: tmiyashita@nature.ca

Dr. Tara Gariepy
Agriculture and Agri-Food Canada
Email: tara.gariepy@agr.gc.ca